

Algebra II with Trigonometry

Chapter 5.6

Study guide

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[4401272_alg-2trig-h-chp-5-6-note-docx.pdf](#)

Below is a complete, self-contained study guide based on your provided document images and extracted text.

1. Short Summary

This document is a lesson on the Law of Cosines, intended for Algebra 2 Trigonometry/Honors students. It focuses on solving oblique triangles (non-right triangles), specifically the Side-Angle-Side (SAS) and Side-Side-Side (SSS) cases. It introduces the Law of Cosines, discusses its use in real-world problems (like navigation and bearings), and shows how to calculate the area of a triangle using Heron's Formula, which can be derived from the Law of Cosines. Several worked examples are provided, covering triangle solving, area calculations, and applications.

2. Core Definitions, Theorems, and Formulas

Law of Cosines (for triangle with sides a , b , c and opposite angles A , B , C):

- $a^2 = b^2 + c^2 - 2bc \cos A$
- $b^2 = a^2 + c^2 - 2ac \cos B$
- $c^2 = a^2 + b^2 - 2ab \cos C$

To Find Angles:

- $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$
- $\cos B = \frac{a^2 + c^2 - b^2}{2ac}$

- $\cos C = \frac{a^2+b^2-c^2}{2ab}$

Bearings (Navigation):

- A bearing is an acute angle measured from due north or due south, indicating direction.

Heron's Formula (Area of a Triangle given sides a, b, c):

- $s = \frac{1}{2}(a + b + c)$ (semiperimeter)
 - $\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$
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3. Worked Study Guide: Step-by-Step Concepts

A. When to Use the Law of Cosines

- Use when you know:
 1. Two sides and the included angle (SAS case)
 2. All three sides (SSS case)
 and want to find an angle

B. Using the Law of Cosines

Step 1: Identify the known values

- Are you solving for a side or an angle? Gather all known sides/angles.

Step 2: Substitute into the right formula

- For a side (ex: find a): $a^2 = b^2 + c^2 - 2bc \cos A$
- Solve for the missing side length.
- For an angle (ex: find A , knowing all sides): $\cos A = \frac{b^2+c^2-a^2}{2bc}$
- Solve for the cosine value, then use $\cos^{-1}$ to find the angle.

Example:

Given $b = 25$, $c = 25$, $A = 140^\circ$ (from Example 1a):

- $a^2 = 25^2 + 25^2 - 2 \times 25 \times 25 \times \cos(140^\circ)$
- Calculate value, then square root for a .

C. Navigation and Bearings

- Always measure the angle from North/South toward East/West.
- Visualize bearings as navigating from a starting point with a certain direction, often requiring triangle-solving.

D. Finding Area with Heron's Formula

Step 1: Calculate the semiperimeter:

- $s = \frac{1}{2}(a + b + c)$

Step 2: Plug into the area formula:

- $\text{Area} = \sqrt{s(s - a)(s - b)(s - c)}$

Example:

Given $a = 3, b = 4, c = 5$:

- $s = \frac{1}{2}(3 + 4 + 5) = 6$
- $\text{Area} = \sqrt{6 \times (6 - 3) \times (6 - 4) \times (6 - 5)} = \sqrt{6 \times 3 \times 2 \times 1} = \sqrt{36} = 6$
units²

E. Solving Real-World Problems

- Set up the triangle based on the scenario (e.g., two trips in different directions, distances as sides, angle between paths).
- Use Law of Cosines to find unknown distances or bearings.
- Use Heron's Formula to find areas, then multiply by unit costs if needed for value.

4. Practice Problems with Answers

Problem 1

In triangle ABC , $a = 7, b = 10, C = 45^\circ$. Find side c .

Answer: $c^2 = 7^2 + 10^2 - 2 \times 7 \times 10 \cos 45^\circ = 49 + 100 - 140 \times 0.7071$
 $= 149 - 98.994 \approx 50.006$ $c \approx 7.07$

Problem 2

Given $a = 9$, $b = 12$, $c = 15$, find angle A (nearest degree).

$$\textbf{Answer: } \cos A = \frac{12^2 + 15^2 - 9^2}{2 \times 12 \times 15} = \frac{144 + 225 - 81}{360} = \frac{288}{360} = 0.8$$
$$A = \cos^{-1}(0.8) \approx 37^\circ$$

Problem 3

Calculate the area of a triangle with sides 13, 14, and 15.

$$\textbf{Answer: } s = (13 + 14 + 15)/2 = 21$$
$$\text{Area} = \sqrt{21(21 - 13)(21 - 14)(21 - 15)} = \sqrt{21 \times 8 \times 7 \times 6} = \sqrt{7056} \approx 84$$

units²

Problem 4

A boat sails 8 miles due east, then 6 miles at a bearing S 30° E. How far is it from its starting point?

$$\textbf{Answer: } \text{Angle between routes} = 180^\circ - 30^\circ = 150^\circ \text{ Use Law of Cosines:}$$
$$d^2 = 8^2 + 6^2 - 2 \times 8 \times 6 \cos 150^\circ \cos 150^\circ = -0.8660$$
$$d^2 = 64 + 36 + 2 \times 8 \times 6 \times 0.8660 \quad d^2 = 100 + 83.136 = 183.136$$
$$d \approx 13.5 \text{ miles}$$

Problem 5

Find the value of a triangular lot with sides 60 ft, 80 ft, 100 ft if each square foot is \$25.

$$\textbf{Answer: } s = (60 + 80 + 100)/2 = 120 \text{ Area} =$$
$$\sqrt{120(120 - 60)(120 - 80)(120 - 100)} = \sqrt{120 \times 60 \times 40 \times 20} = \sqrt{576,000}$$

ft² Value = 759 × \$25 = \$18,975

5. Assumptions and Ambiguities

- The sides and angles referred to follow the convention that side a is opposite A , side b opposite B , etc.

- Bearings are measured as stated, from North or South toward East or West, as illustrated.
 - Navigation and area problems use Law of Cosines unless right triangle methods are applicable (not the case here).
 - Rounding in numerical answers is as typical for trigonometry problems (2 decimal places unless otherwise specified).
 - Some diagrams/information in examples are not fully clear in text, but core method and approach have been followed as per standard mathematics.
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This study guide lets you review the Law of Cosines and related applications independently.